Pavithra PM Nair

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Summary

AI researcher working on building responsible AI, with a focus on evaluating large language models for fairness and cultural robustness. My work includes auditing LLMs for demographic and cross-lingual biases in high-stakes domains such as legal judgment and economic advice, studying how AI assistance shapes human decision-making, and resolving ambiguity in natural language requirements. I build systems and run empirical studies aimed at making language technologies more reliable, fair, and genuinely useful across linguistic and cultural contexts.

Education

Amrita Vishwa Vidyapeetham
Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence) *2020 - 2024*
CGPA - 9.36/10, Department Rank - 6

Ben-Gurion University of the Negev
Data Mining and Business Intelligence Summer School *August 2023 - September 2023*

Experience

Amrita Vishwa Vidyapeetham
Scientist *April 2026 - Present*

- **Team Leadership** – Leading a team of 3 researchers studying fairness, reliability, and cross-lingual behavior of large language models.
- **Bias Auditing at Scale** – Created RupeeBias, a counterfactual audit of demographic bias in LLM economic advice (∼39,000 prompts, 87 India-specific identifiers, English and Hinglish), showing advice shifts ∼20% under demographic perturbation alone; submitted to NeurIPS Evaluation & Datasets Track 2026.

Tata Research Development and Design Centre (TCS Research)
Researcher *August 2024 - April 2026*

- **Ambiguity Handling** - Developed a novel RAG-based approach for detecting and resolving pragmatic ambiguities in natural language requirements using multi-level domain knowledge bases; published at EASE 2026.
- **Regulatory Compliance Automation** - Built Reg2Req, an automated LLM pipeline that identifies requirement-bearing clauses in regulations and derives traceable, plain-language software requirements, evaluated on the full GDPR and EU AI Act (RE 2026 Industrial Innovation Track); co-developed C-PASS, an organization-centric framework for compliance and penalty assessment (Information and Software Technology).

Prodago - AI Governance (as contracted through Traboda Cyberlabs)
Researcher *November 2023 - August 2024*

- **RAG-based Compliance Chatbot** - Built a retrieval-augmented generation pipeline (Pinecone) for a compliance chatbot, improving output quality by 28% through systematic prompt engineering, and established Ragas-based evaluation workflows to guide model selection.

Association for Computing Machinery
Research Intern *July 2023 - August 2023*

- **Bias Detection & Mitigation** - Analyzed demographic biases in computer vision datasets and contributed to a mitigation approach combining data augmentation and reweighing, improving model fairness without sacrificing accuracy.

Publications

1. **Pavithra PM Nair**, Gilad Gressel, Krishnashree Achuthan. **"Sorry, Can't Help You": How Large Language Models Judge Failures to Help Across Languages.** *4th Workshop on Cross-Cultural Considerations in NLP (C3NLP), ACL 2026.* **Best Paper Award.**
2. **Pavithra PM Nair**, Preethu Rose Anish. **From Regulation to Requirements: An Automated Requirement Derivation and Explanation Pipeline.** *34th IEEE International Requirements Engineering Conference (RE), Industrial Innovation Track, 2026.*
3. **Pavithra PM Nair**, Preethu Rose Anish. **Detecting and Resolving Pragmatic Ambiguities in Natural Language Requirements Using Multi-Level Domain Knowledge Bases.** *International Conference on Evaluation and Assessment in Software Engineering (EASE), 2026.*

4. **Pavithra PM Nair**, Preethu Rose Anish. **Vichara: Appellate Judgment Prediction and Explanation for the Indian Judicial System.** *Artificial Intelligence and Law Workshop, AAAI 2026.*
5. Gokul Rejithkumar, Sachin Pawar, **Pavithra PM Nair**, Nitin Ramrakhiyani, Preethu Rose Anish. **C-PASS: An Organization-centric Framework for Compliance and Penalty Assessment Using Large Language Models.** *Information and Software Technology, 2026.*
6. **Pavithra PM Nair**, Gilad Gressel, Krishnashree Achuthan. **Lost in Translation? How Language Shapes Responsibility Attribution in Large Language Models.** *4th Workshop on Cross-Cultural Considerations in NLP (C3NLP), ACL 2026.*
7. Srividya Sheshadri, Manju Balakrishna Pillai, Renuka Kumar, **Pavithra PM Nair**, Apoorv Gupta, Megha TM, Sooraj S Nair, Balu Mohandas Menon, Unnikrishnan Radhakrishnan, Bhavani Rao R. **Sruthi: What Becomes Sayable Through Peer-Mode Reflective Scaffolding in Fieldwork Communication.** *ACM SIGCHI Conversational User Interfaces (CUI), 2026.*
8. **Pavithra PM Nair**, Bhavik Talaviya, Shourya Bhushan, Rahul Pankajakshan, Seema Guruvadoo, Avinash Agarwal, Gilad Gressel, Krishnashree Achuthan. **RupeeBias: Auditing Demographic Bias in Indian Economic Guidance from Large Language Models.** *Under Review, 2026.*
9. **Pavithra PM Nair**, Gilad Gressel, Malavika Nambrath Anand, Krishnashree Achuthan. **To Solicit or Not to Solicit? Impact of AI Assistance Delivery Mechanisms on Decision-Making.** *International Journal of Human–Computer Interaction, 2025.*
10. Akhbar Sha, Anvita Reddy Inture, Jose Joseph, **Pavithra PM Nair**, MJ Sheeba, T Anjali. **An Integration of Indian Philosophy and Machine Learning for Human-like Cognition.** *International Conference on Automation, Computing and Renewable Systems (ICACRS), 2023.*

Projects

Athena – AI Socratic Chatbot for Cybersecurity Education Funded by Patrick J. McGovern Foundation
Researcher & Developer *Jan 2024 - August 2024*

- **AI Chatbot Development** – Designed and developed an LLM-powered chatbot using Python and LangChain, tailored to stimulate critical thinking and cognitive engagement through Socratic questioning techniques.
- **LLM Performance Optimization** – Improved response quality through prompt engineering and RAG, leading to a tangible increase in user engagement and knowledge retention.
- **User Testing and Platform Integration** – Integrated the chatbot into the Traboda Cyberlabs Capture the Flag (CTF) platform and conducted A/B testing to measure its effectiveness in engaging users and promoting deeper learning.

Extracurriculars, Leadership and Initiatives

amFOSS - Computer Science Club
Mentor & Member *2020 - 2024*

- **Open Source Contribution** - Contributed to open source projects both within the club (amFOSS CMS) and external projects (Navidrome, Ersilia Open Source Initiative).
- **Podcast** - Led ideation for and co-hosted the amFOSS Podcast to promote open-source engagement.

Google Developer Student Club
Lead *2023 - 2024*

- **Technical Knowledge Sharing** – Conducted hands-on workshops on topics such as web development, AI, and cloud computing, emphasizing practical skills and real-world applications.

Awards and Honors

- **Best Paper Award, C3NLP Workshop at ACL 2026**: Awarded for "Sorry, Can't Help You: How Large Language Models Judge Failures to Help Across Languages" at the 4th Workshop on Cross-Cultural Considerations in NLP.
- **Scholarship Recipient, Data Mining and Business Intelligence Summer School, Ben-Gurion University of the Negev (2023)**: Awarded a full scholarship to attend the competitive summer school program at BGU, focusing on machine learning and cybersecurity.
- **Economic Times Campus Star (2023)**: Recognised by The Economic Times after a four-round evaluation process, including technical assessments and interviews, for demonstrating technical expertise and professional skills.
- **Academic Achievement Award, Amrita Vishwa Vidyapeetham (2023)**: Recognized for outstanding academic performance in the B.Tech. Computer Science and Engineering (Artificial Intelligence) program.
- **Finalist, European Rover Competition (2021)**: Contributed to the software enabling remote navigation of a rover through a simulated Mars-like environment.